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(54) **CONNECTOR**

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CPC ***H01R 13/62977*** (2013.01)

(57) **ABSTRACT**

A connector adapted to mate with a mating connector includes an outer housing, a slider, and an operation lever. The outer housing is adapted to receive the mating connector, and includes a catching portion. The slider is provided in the outer housing and is slidingly movable in a predetermined direction. The slider includes a catching portion adapted to deflect at a predetermined position and to be caught by the catching portion of the outer housing. The operation lever is combined with the slider and is adapted to be turned between a pre-turning first position and a post-turning second position. The slider is slidingly movable with the turning of the operation lever. The catching portion of the slider and the operation lever in the first position can be caught on the caught portion of the outer housing.

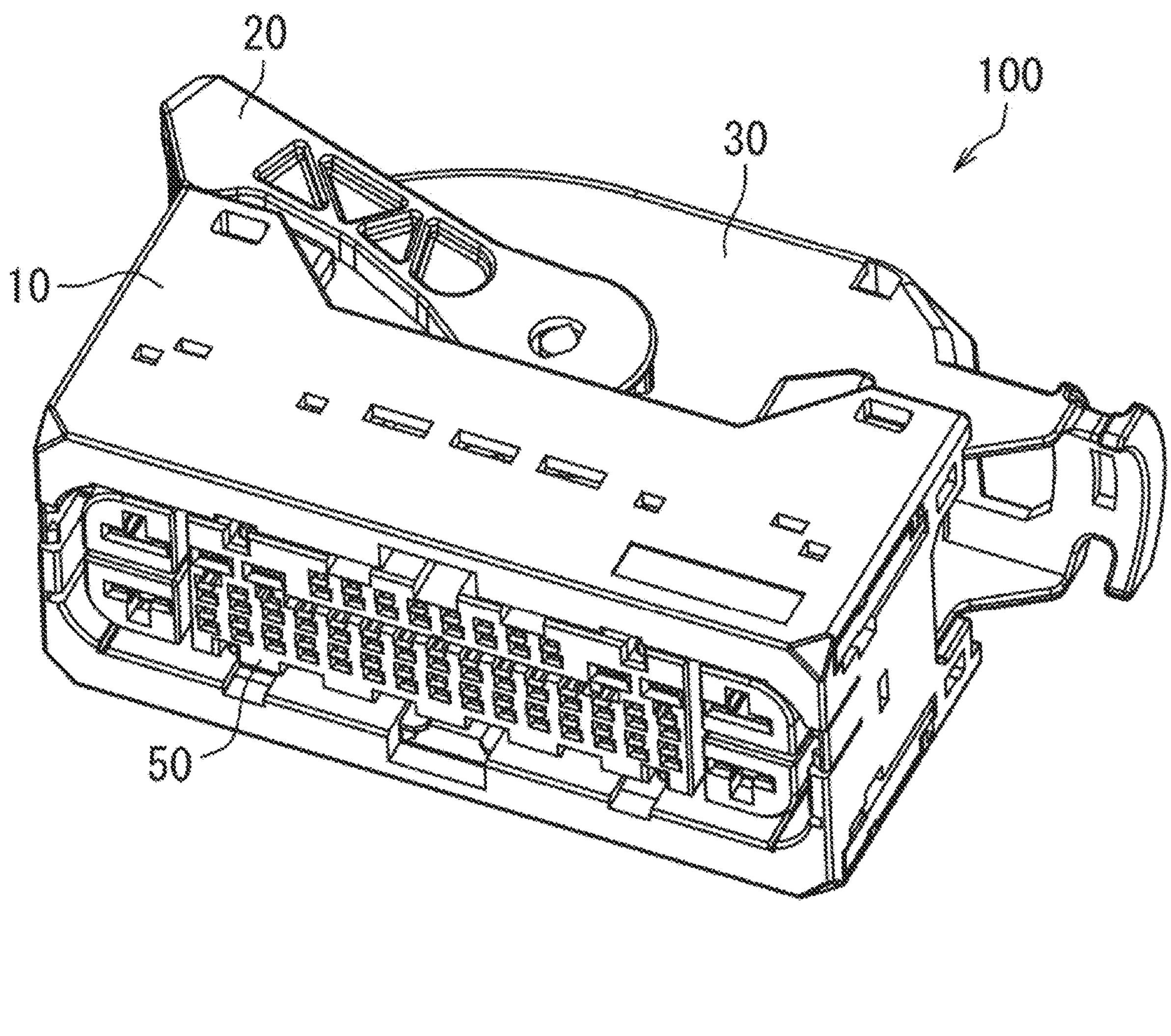


FIG. 1

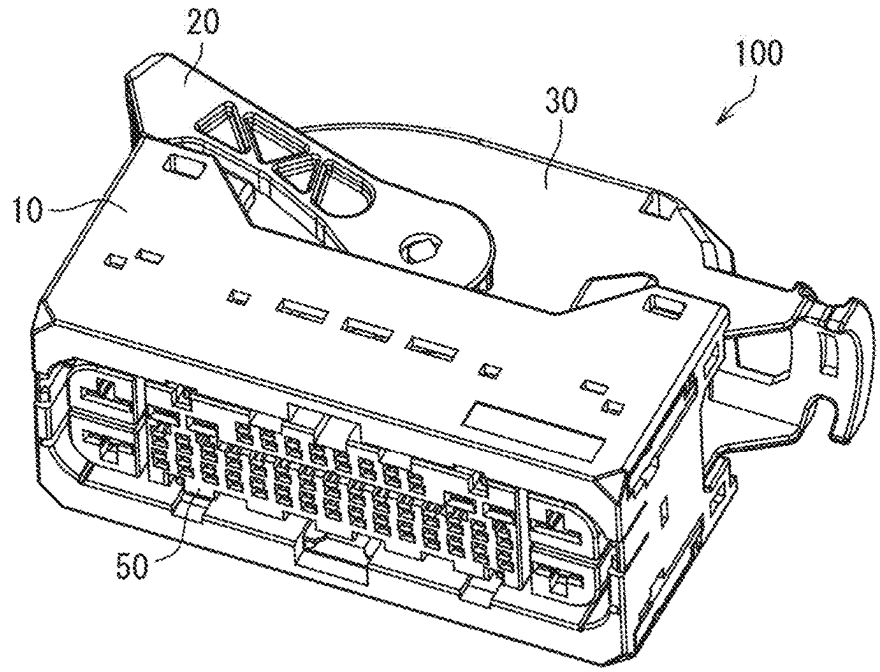


FIG. 2

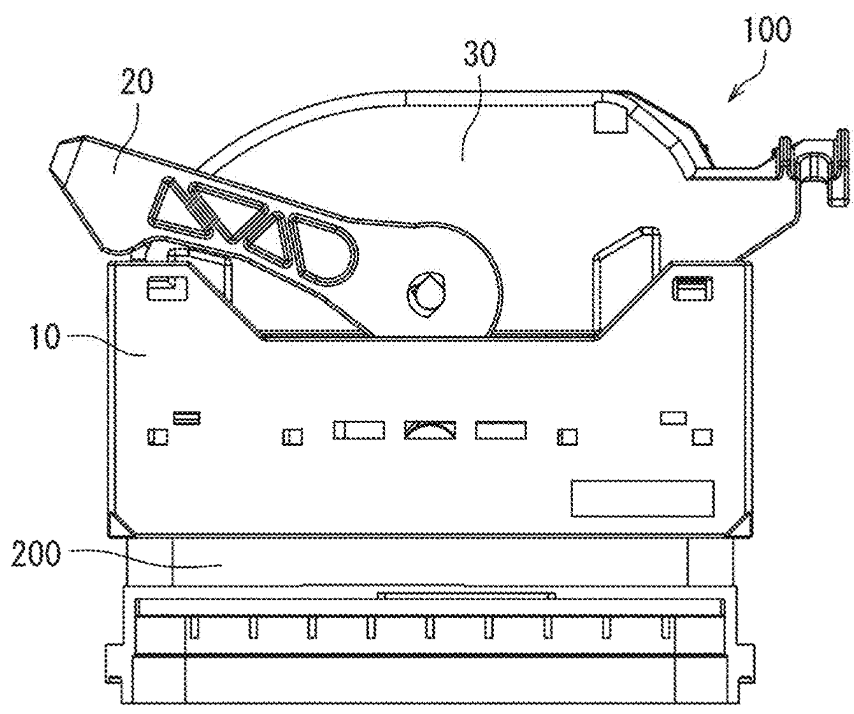


FIG. 3

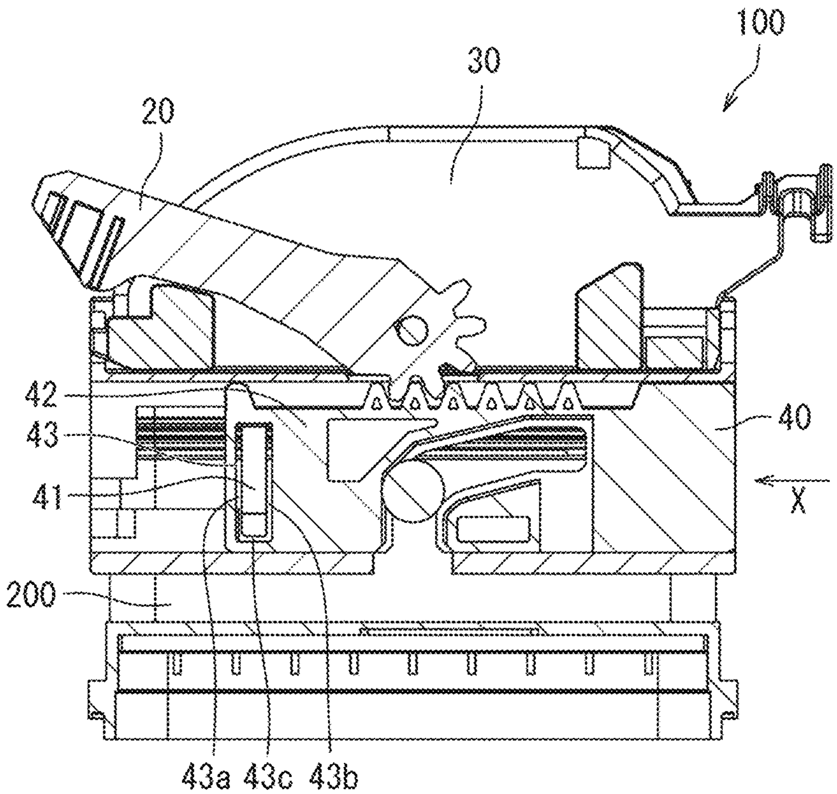


FIG. 4A

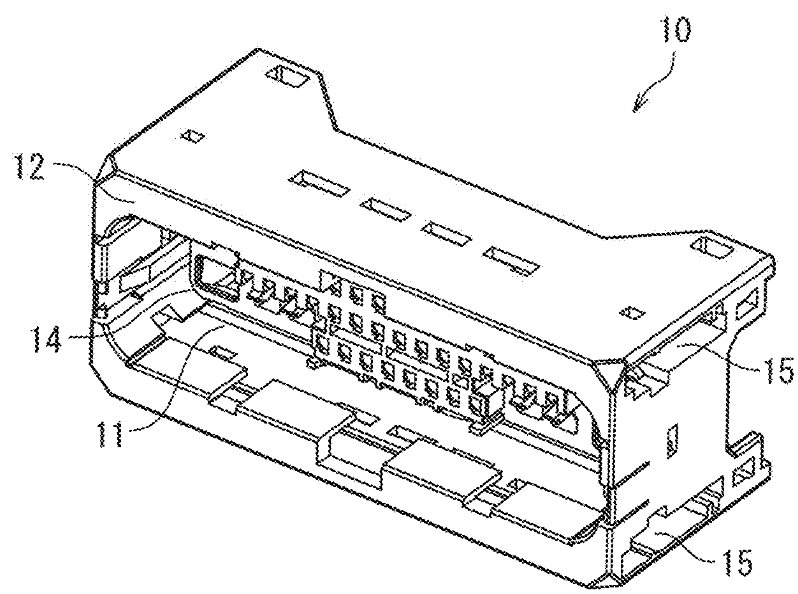


FIG. 4B

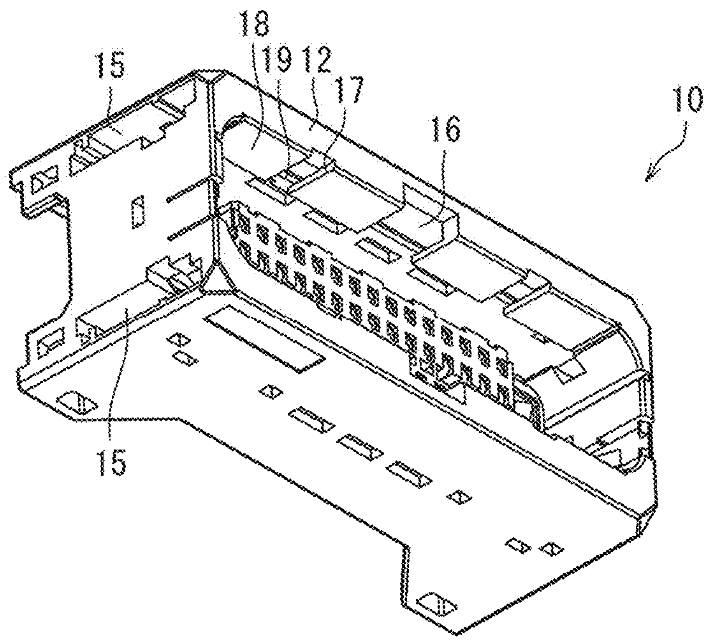


FIG. 4C

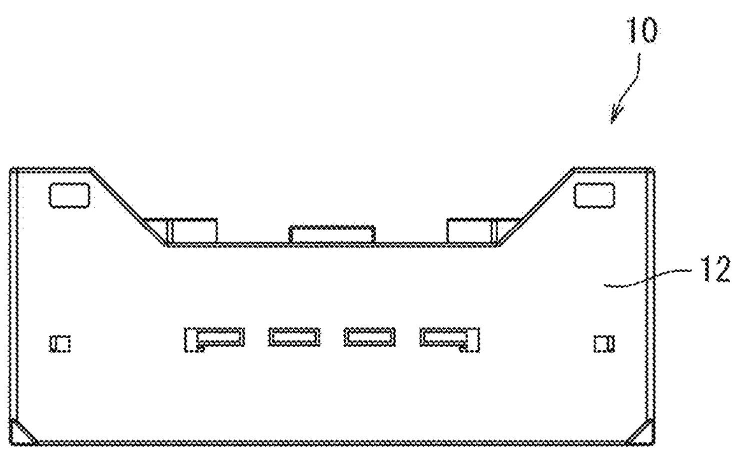


FIG. 4D

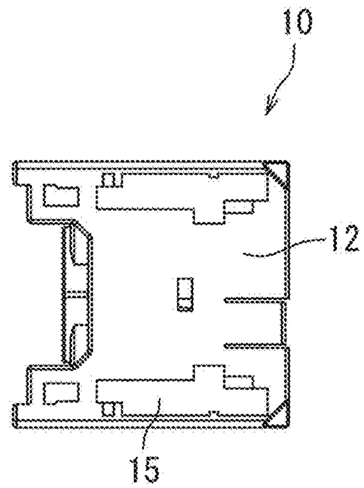


FIG. 4E

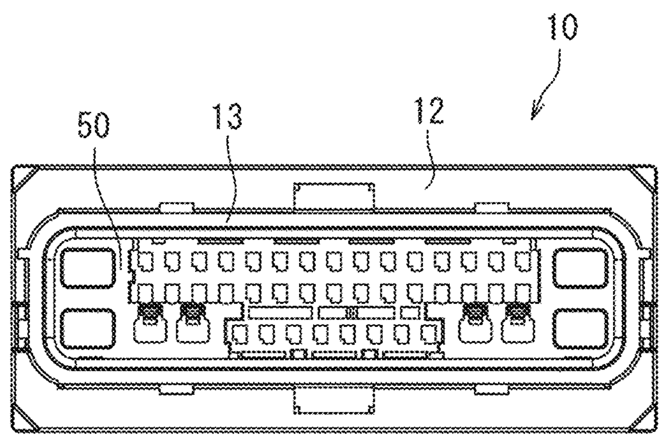


FIG. 5

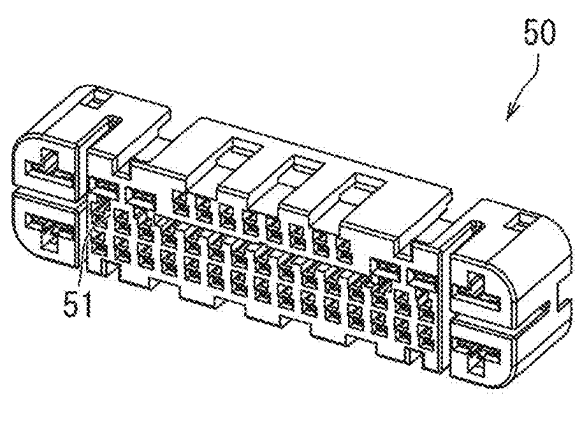


FIG. 6A

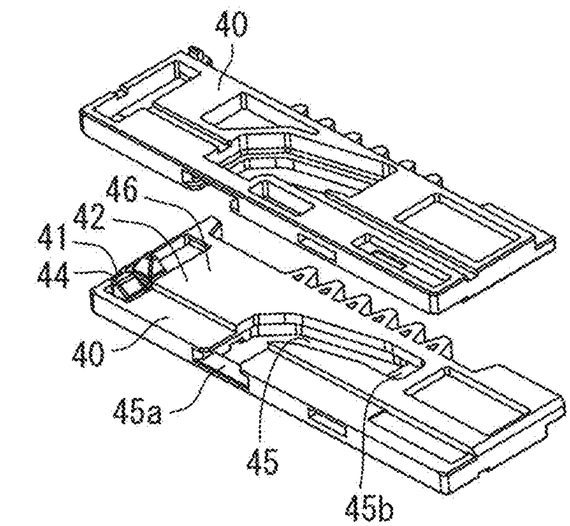


FIG. 6B

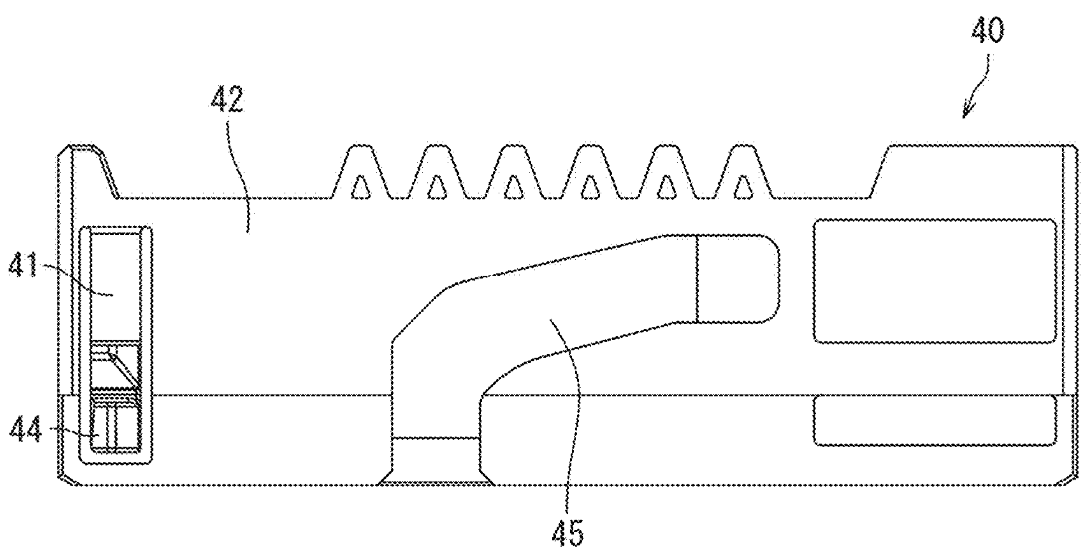


FIG. 6C

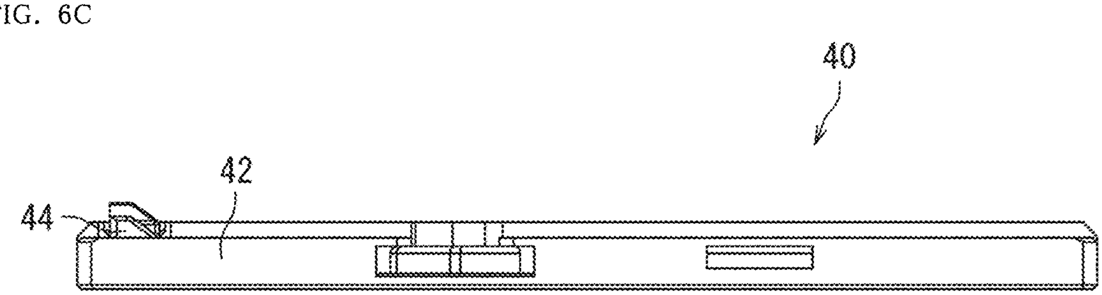


FIG. 6D

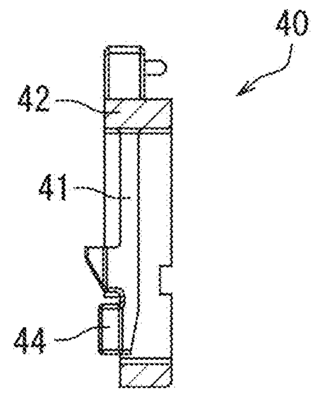


FIG. 7

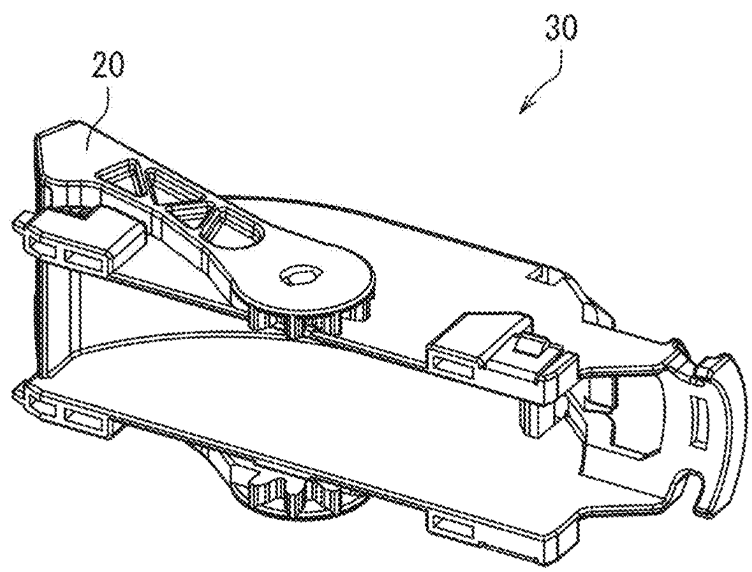


FIG. 8

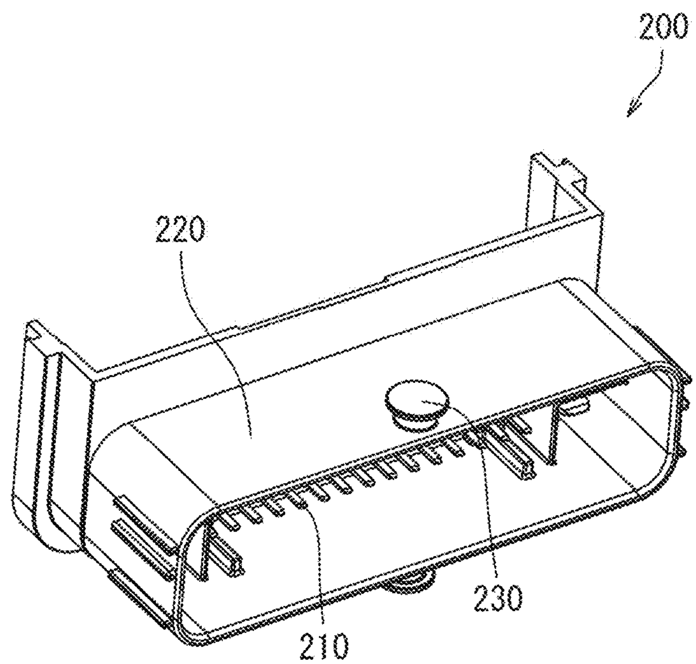


FIG. 9

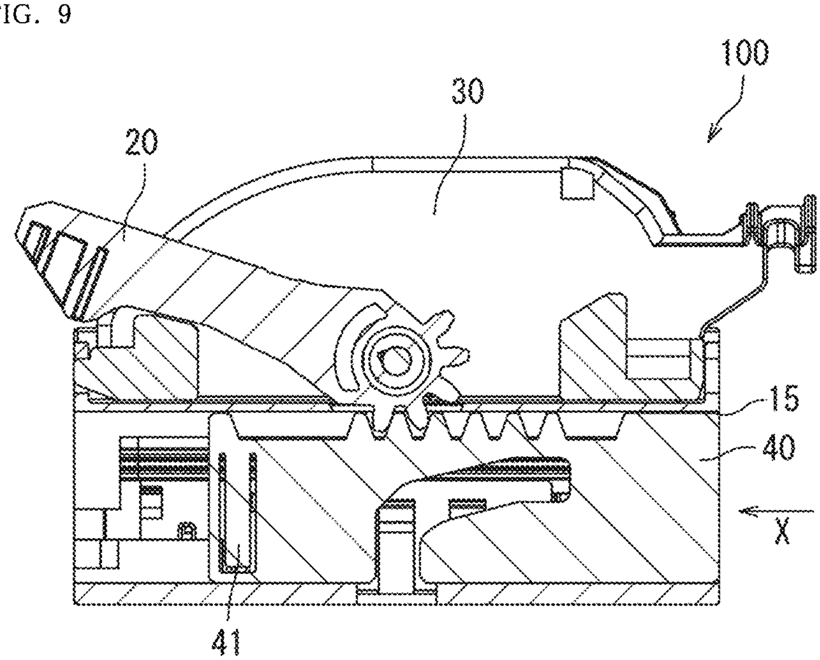


FIG. 10

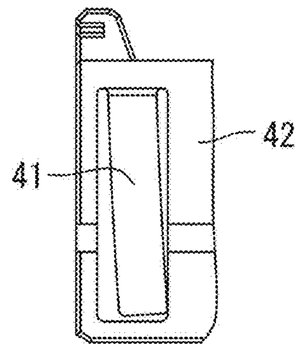


FIG. 11

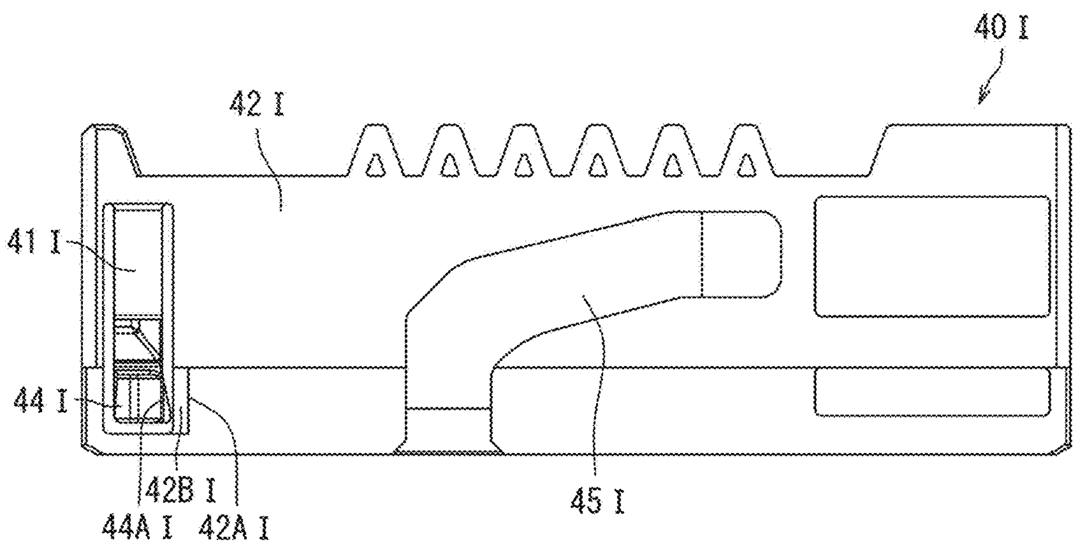


FIG. 12

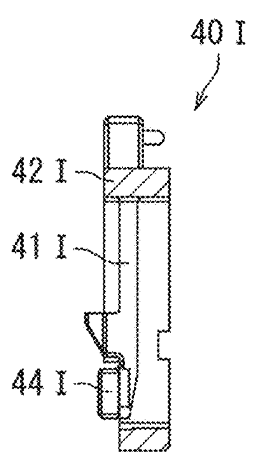


FIG. 13

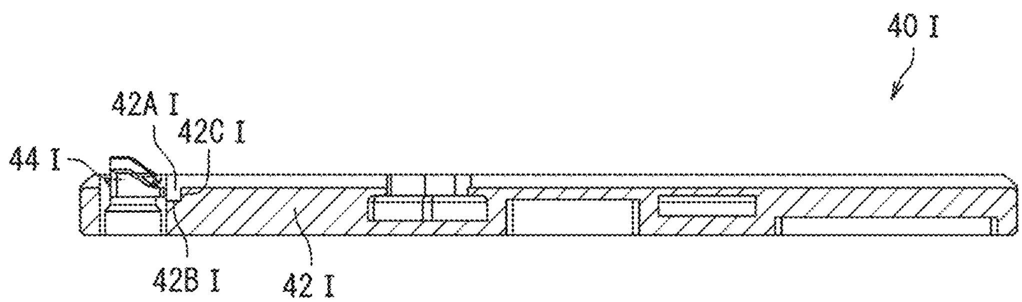
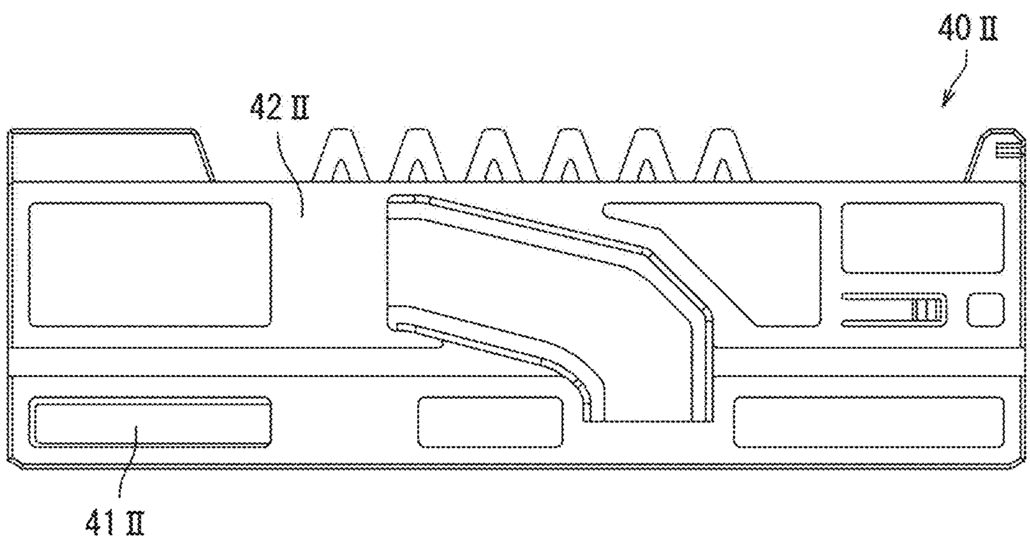


FIG. 14



CONNECTOR

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of Japanese Patent Application No. 2024-030155 filed on Feb. 29, 2024, the whole disclosure of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a connector. Specifically, the present disclosure relates to a connector having a lever mechanism.

BACKGROUND OF THE INVENTION

[0003] In the prior art, a connector including a lever mechanism for reducing a force required to mate with a mating connector. Such a connector includes a housing, a sliding member (equivalent to a slider), and an operation lever for slidingly moving the slider. Turning operation of this operation lever causes a cam pin of the mating connector to move along a cam groove formed in the slider, thereby mating the connector and the mating connector.

[0004] A connector having a conventional lever structure has a problem to be overcome. Specifically, during a mating operation of the connector having such a lever structure as mentioned above, with the operation lever in an initial position (that is, a pre-turning position of the operation lever), the connector and the mating connector are combined. In the connector of the prior art, the operation lever is retained by means of a protrusion provided on a cover. That is, the operation lever is sufficiently retained by means of such a protrusion, and therefore, if load is applied to the operation lever, the operation lever may be unintentionally detached from the protrusion and displaced from the initial position, resulting in a reduction in working efficiency in the mating operation.

[0005] The present disclosure has been made in view of such a problem. Therefore, an object of the present disclosure is to provide a connector having a more suitable operation lever retaining mechanism.

SUMMARY OF THE INVENTION

[0006] According to an embodiment of the present disclosure, a connector adapted to mate with a mating connector includes an outer housing, a slider, and an operation lever. The outer housing is adapted to receive the mating connector, and includes a catching portion. The slider is provided in the outer housing and is slidingly movable in a predetermined direction. The slider includes a catching portion adapted to deflect at a predetermined position and to be caught by the catching portion of the outer housing. The operation lever is combined with the slider and is adapted to be turned between a pre-turning first position and a post-turning second position. The slider is slidingly movable with the turning of the operation lever. The catching portion of the slider and the operation lever in the first position can be caught on the caught portion of the outer housing.

DRAWINGS

[0007] The accompanying drawings incorporated therein and forming a part of the specification illustrate the present disclosure and, and together with the description, further serve to explain the principles of the disclosure and to enable those skilled in the relevant art to manufacture and use the embodiments described herein.

[0008] FIG. **1** is an isometric view schematically showing a connector of the present disclosure with an operation lever in a first position;

[0009] FIG. **2** is a front view schematically showing a connector of the present disclosure in a temporarily mated state with a mating connector with the operation lever in the first position and;

[0010] FIG. **3** is a front view schematically showing a connector of the present disclosure (an outer housing which is a component thereof is not shown) in the temporarily mated state with the mating connector with the operation lever in the first position;

[0011] FIG. **4**A is an isometric view schematically showing the outer housing of the connector of the present disclosure;

[0012] FIG. **4**B is an isometric view schematically showing the outer housing of the connector of the present disclosure as seen from a different angle from FIG. **4**A;

[0013] FIG. **4**C is a front view schematically showing the outer housing of the connector of the present disclosure;

[0014] FIG. **4**D is an end view schematically showing the outer housing of the connector of the present disclosure;

[0015] FIG. **4**E is a top view schematically showing the outer housing of the connector of the present disclosure;

[0016] FIG. **5** is an isometric view schematically showing an inner housing of the connector of the present disclosure;

[0017] FIG. **6**A is an isometric view schematically showing a slider of the connector of the present disclosure;

[0018] FIG. **6**B is a front view schematically showing an inner main face of the slider of the connector of the present disclosure;

[0019] FIG. **6**C is a bottom view schematically showing the slider of the connector of the present disclosure;

[0020] FIG. **6**D is a cross-sectional view schematically showing the slider of the connector of the present disclosure as seen from one end face side;

[0021] FIG. **7** is an isometric view schematically showing a cover with the operation lever of the connector of the present disclosure pivotally supported thereon;

[0022] FIG. **8** is an isometric view schematically showing the mating connector mateable with the connector of the present disclosure;

[0023] FIG. **9** is a front view schematically showing the connector of the present disclosure (the outer housing which is a component thereof is not shown) at the time of unexpected turning of the operation lever before temporary mating with the mating connector;

[0024] FIG. **10** is a partially enlarged front view schematically showing a behavior of a catching portion of the slider which is a component of the connector of the present disclosure at the time of unexpected turning of the operation lever before temporary mating with the mating connector;

[0025] FIG. **11** is a front view schematically showing an inner main face of an alternative example of the slider of the connector of the present disclosure;

[0026] FIG. **12** is a cross-sectional view schematically showing the alternative example of the slider of the connector of the present disclosure as seen from one end face side;

[0027] FIG. **13** is a cross-sectional view schematically showing the alternative example of the slider of the connector of the present disclosure as seen from a bottom side; and

[0028] FIG. **14** is a front view schematically showing an inner main face of a further alternative example of the slider of the connector of the present disclosure.

[0029] The features disclosed in this disclosure will become more apparent in the following detailed description in conjunction with the accompanying drawings, where similar reference numerals always identify the corresponding components. In the accompanying drawings, similar reference numerals typically represent identical, functionally similar, and/or structurally similar components. Unless otherwise stated, the drawings provided throughout the entire disclosure should not be construed as drawings drawn to scale.

DETAILED DESCRIPTION

[0030] Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein the like reference numerals refer to the like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiment set forth herein; rather, these embodiments are provided so that the present disclosure will be thorough and complete, and will fully convey the concept of the disclosure to those skilled in the art.

[0031] In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

[0032] A connector according to an embodiment of the present disclosure will be specifically described below with reference to the drawings. Each element in the drawings is schematically and illustratively shown merely for describing the present disclosure, and may have a different exterior, dimensional ratio, and/or the like from an actual one.

[0033] First, in order to grasp a general structure of the connector of the present disclosure, a basic configuration of the connector of the present disclosure will be described below with reference to the drawings.

[0034] FIG. **1** is an isometric view schematically showing the connector of the present disclosure with an operation lever in a first position. FIG. **4**A is an isometric view schematically showing an outer housing of the connector of the present disclosure. FIG. **4**B is an isometric view schematically showing the outer housing of the connector of the present disclosure as seen from a different angle from FIG. **4**A. FIG. **4**C is a front view schematically showing the outer housing of the connector of the present disclosure. FIG. **4**D is an end view schematically showing the outer housing of the connector of the present disclosure. FIG. **4**E is a top view schematically showing the outer housing of the connector of the present disclosure. FIG. **5** is an isometric view schematically showing an inner housing of the connector of the present disclosure.

[0035] FIG. **6**A is an isometric view schematically showing a slider of the connector of the present disclosure. FIG. **6**B is a front view schematically showing an inner main face of the slider of the connector of the present disclosure. FIG. **6**C is a cross-sectional view schematically showing the slider of the connector of the present disclosure. FIG. **6**D is a cross-sectional view schematically showing the slider of the connector of the present disclosure as seen from one end face side. FIG. **7** is an isometric view schematically showing a cover with the operation lever of the connector of the present disclosure pivotally supported thereon. FIG. **8** is an isometric view schematically showing a mating connector matable with the connector of the present disclosure. FIG. **9** is a front view schematically showing the connector of the present disclosure (the outer housing which is a component thereof is not shown) at the time of unexpected turning of the operation lever before temporary mating with the mating connector. FIG. **10** is a partially enlarged front view schematically showing a behavior of a catching portion of the slider which is a component of the connector of the present disclosure at the time of unexpected turning of the operation lever before temporary mating with the mating connector.

[0036] A connector **100** of the present disclosure includes, as main components, an outer housing **10**, a slider provided in the outer housing **10**, and an operation lever **20** combined with the slider and so attached as to be turnable or rotatable between a pre-turning first position shown in FIG. **1** and a post-turning second position.

[0037] The slider **40** slidingly moves in a predetermined direction in the outer housing **10** as the operation lever **20** is turned. That is, an operator can slidingly move the slider **40** by turning the operation lever **20** (see FIG. **6**A and the like). The operation lever **20** helps to mate the connector **100** of the present disclosure with a mating connector **200** by turning operation.

[0038] In addition, the slider **40** has, on an inner main face **46** side, a cam groove **45** relative to which a cam pin **230** (see FIG. **8**) of the mating connector **200**, which will be described later, can move during the sliding movement in the predetermined direction mentioned above.

[0039] In the present disclosure, when the cam pin **230** is located at a first end **45*a*** as an inlet end of the cam groove **45**, temporary mating of the mating connector **200** with the connector **100** can be achieved. That is, a state of "temporary mating of a mating connector" herein refers to a provisional (incomplete) mated state of the mating connector **200** with the connector **100** of the present disclosure. Specifically, the state of "temporary mating of a mating connector" refers to a state where the operation lever **20** is in the first position, and simultaneously the cam pin **230** of the mating connector is located at the first end **45*a*** as the inlet end of the cam groove **45** of the slider **40**. In other words, a state where the operation lever **20** is in the first position can be a state where the cam pin **230** of the mating connector is located at the first end **45*a*** as the inlet end of the cam groove **45** of the slider **40**.

[0040] In addition, when the cam pin **230** is located at a terminal second end **45*b*** opposite to the first end of the cam groove **45**, final mating of the mating connector **200** with the connector **100** of the present disclosure can be achieved. That is, a state of "final mating of the mating connector" herein refers to a state where the connector **100** of the present disclosure and the mating connector **200** are completely mated with each other. Specifically, the state of "final mating of the mating connector" refers to a state where the operation lever **20** is in the second position, that is, turning of the same lever **20** is completed, and simultaneously the cam pin **230** of the mating connector is located at the second end **45**_b_ of the cam groove **45** of the slider **40**. In other words, the state where the operation lever **20** is in the second position can be a state where the cam pin **230** of the mating connector is located at the second end **45**_b_ of the cam groove **45** of the slider **40**.

[0041] The outer housing **10** can be configured to be capable of accommodating an entire inner housing **50** and part of the mating connector **200**. The outer housing **10** has a first inner space **14** with a first opening portion **11** in a direction of mating with the mating connector **200**. The first inner space **14** is formed by a body portion **12** of the outer housing **10**.

[0042] Through this first opening portion **11**, the inner housing **50** and the mating connector **200** can be inserted and arranged in the first inner space **14**. The inner housing **50** has a plurality of through portions **51** in the mating direction of the mating connector **200** accommodating male contacts **210** (see FIG. **8**) of the mating connector **200** and contactable with the same contacts.

[0043] In addition, in the first inner space **14** of the outer housing **10**, with the inner housing **50** arranged therein, a continuous space **13** can be provided between the body portion **12** of the outer housing **10** and the inner housing **50**. This continuous space **13** can be a space for inserting an enclosing portion **220** accommodating the male contacts **210** of the mating connector **200** mentioned above (see FIG. **4**E). During mating of the mating connector **200**, the enclosing portion **220** of the mating connector **200** is located in this continuous space **13**.

[0044] The body portion **12** of the outer housing **10** has, therein, a second inner space **16** with a second opening portion **15** in a direction intersecting the mating direction of the mating connector **200** mentioned above, for example in a direction perpendicular thereto. Two second inner spaces **16** can be provided in the body portion **12** of the outer housing **10**.

[0045] The slider **40** (see FIGS. **6**A to **6**D) can be inserted through such a second opening portion **15**, and retained in the second inner space **16**. Herein, this second inner space **16** can also be referred to as a "slider retaining space". In addition, the inserted slider **40** is slidingly movable in the second inner space **16** in the predetermined direction, for example a direction perpendicular to the mating direction of the mating connector **200** mentioned above.

[0046] A cover **30** can be attached to the body portion **12** of the outer housing **10** from a direction opposite to the mating direction of the mating connector **200** mentioned above, that is, from a side opposite to the first opening portion **11**.

[0047] The cover **30** can draw and align electrical wires connected to a plurality of terminals plugged in the connector of the present disclosure in a predetermined routing direction. The operation lever **20** can extend across the cover **30** along a short-side direction of the connector, and be pivotally supported on opposite sides of the cover **30** so as to be turnable or rotatable.

[0048] The connector **100** of the present disclosure may include a retainer plugged in the inner housing **50**. The retainer may position and secure terminals of the mating connector in the inner housing **50**. In addition, the connector **100** of the present disclosure may further include a seal material for waterproofing. For example, the connector **100** may include the seal material on an inner face and/or an outer periphery of the inner housing **50**. The seal material may waterproof the terminals in the connector, and shut off water between the inner housing **50** and the mating connector **200**, for example.

[0049] In addition, the components such as a housing such as the outer housing **10**, the slider **40**, the operation lever **20**, and the cover **30** may be formed from an electrically insulating non-conductive material. These electrically insulating components may include a resin material having an electrically insulating property. Such electrically insulating components may include, but be not limited to, at least one kind of thermosetting resin selected from the group consisting of epoxy resins, phenolic resins, silicone resins, and unsaturated polyester resins, for example. In addition, members different from each other may include respective different resin materials.

[0050] Having the configuration described above as a premise, the present disclosure is characterized by the following points.

[0051] Specifically, the slider **40** mentioned above has a catching portion **41** having deflectability at a predetermined spot, and the outer housing **10** has a caught portion **17** on which the catching portion **41** of the slider **40** can be caught. In an example, as shown in FIGS. **3**, **6**B, and the like, the catching portion **41** of the slider **40** can be configured to have its long sides extending in a direction substantially perpendicular to a moving direction X of the slider **40**.

[0052] The catching portion **41** of the slider **40**, with the operation lever **20** in the first position mentioned above, can be caught on the caught portion **17** of the outer housing **10**. Then, temporary mating of the mating connector **200** causes the mating connector **200** to abut against the catching portion **41** of the slider **40**, allowing the catching portion **41** to deflect, thereby enabling the slider **40** to be released from a caught state on the caught portion **17** of the outer housing **10**.

[0053] Having such a feature enables the catching portion **41** of the slider **40** in the caught state on the caught portion **17** of the outer housing **10** to be retained before the temporary mating of the mating connector **200**. That is, the operation lever **20** can be retained in the pre-turning first position.

[0054] On the other hand, the release of the catching portion **41** of the slider **40** from the caught state during the temporary mating of the mating connector **200** enables the operation lever **20** to be turned from the pre-turning first position toward the post-turning second portion to slidingly move the slider in the predetermined direction. In the present disclosure, when such a sliding movement of the slider **40** is completed, final mating of the mating connector **200** with the connector **100** of the present disclosure becomes possible.

[0055] Also from the above, in the present disclosure, unless the temporary mating of the mating connector **200** is done, the abutment of the mating connector **200** against the catching portion **41** of the slider **40** and the subsequent deflection of the catching portion **41** to release the catching portion **41** of the slider **40** from the caught state do not occur. Therefore, even if an external force to turn the operation lever **20** from the pre-turning first position toward the post-turning second position acts before the temporary mating of the mating connector **200**, the catching portion **41** of the slider **40** in the caught state on the caught portion **17** of the outer housing **10** is retained, so that the operation lever **20** can be retained in the pre-turning first position (see FIG. **9**). For this reason, according to the connector **100** of the present disclosure, a more suitable operation lever retaining mechanism can be provided.

[0056] An example of the catching portion **41** of the slider **40** having deflectability mentioned above will be described below. The catching portion **41** is provided inside a slit **43** formed in a body portion **42** of the slider **40** (see FIGS. **3**, **6**A, and the like). Specifically, the catching portion **41** having deflectability can be provided by providing a first slit **43***a* and a second slit **43***b* opposite to each other and a third slit **43***c* connecting these two slits **43***a*, **43***b* in the body portion **42** of the slider. A protrusion **44** is provided at an end of this catching portion **41**.

[0057] In addition, the outer housing **10** mentioned above has an inner wall portion **18** forming part of the second inner space **16** for retaining the slider **40**. This inner wall portion **18** has an opening portion **19** in which the protrusion **44** of the above catching portion **41** can be located.

[0058] In the present disclosure, with the operation lever **20** in the pre-turning first position, the protrusion **44** of the catching portion **41** can be caught on a catching wall forming the opening portion **19** of the outer housing **10**. This avoids a sliding movement of the slider **40**, and the pre-turning first position of the operation lever **20** accompanying the movement of the slider **40** is retained.

[0059] Further, in the case where the deflectable catching portion **41** is provided by forming the slit **43**, the mating connector **200** can abut against the protrusion **44** of the catching portion **41** during the temporary mating of the mating connector **200**.

[0060] Such abutment causes the catching portion **41** to deflect, since the catching portion **41** has deflectability, enabling the protrusion **44** of the catching portion **41** to be released from being caught on the catching wall forming the opening portion **19** of the outer housing **10**. This makes the slider **40** slidingly movable in the predetermined direction, thereby enabling the operation lever **20** to be turned from the pre-turning first position toward the post-turning second position.

[0061] Further, in the present disclosure, the second inner space **16** as the slider retaining space locally has a catching portion accommodating space at an end side thereof in which the catching portion **41** of the slider **40** after completion of the sliding movement can be accommodated. Since the catching portion **41** is accommodated in such a catching portion accommodating space, the slider **40** can avoid slidingly moving again in a direction opposite to the predetermined direction mentioned above. This suitably enables the operation lever **20** to avoid returning from the post-turning second position to the pre-turning first position after the operation lever **20** is turned.

[0062] In the present disclosure, it is preferred that the connector **100** take the following aspect. As described above, in the present disclosure, before the temporary mating of the mating connector **200**, the catching portion **41** of the slider **40** is caught on the caught portion **17** of the outer housing **10**. Specifically, the protrusion **44** of the catching portion **41** can be caught on the catching wall forming the opening portion **19** (see FIG. **4**B) of the outer housing **10**.

[0063] In this point, before the temporary mating of the mating connector **200**, due to the catching portion **41** having deflectability, distortion (or twist) may occur in the catching portion **41**, causing the catching portion **41** to be unexpectedly released from being caught on the caught portion **17**.

[0064] In view of this point, as shown in FIG. **11**, it is preferred that a catching portion **41**I in a front view have a local inclined portion **44**AI on a side of a protrusion **44**I thereof, and that an inner main face of a body portion **421** of a slider **40**I locally have a cutout portion **42**AI abuttable against this local inclined portion **44**AI.

[0065] This cutout portion **42**AI, due to its form, has a first cutout face **42**BI extending in the moving direction of the slider **40**, and a second cutout face **42**CI extending in a direction substantially perpendicular to the moving direction of the slider **40**.

[0066] By taking such a configuration, even if distortion occurs in the deflectable catching portion **41**, causing a force by which the catching portion **41** is unexpectedly released from being caught on the caught portion **17**, the local inclined portion **44**AI of the catching portion **41**I can abut against both the first cutout face **42**BI extending in the moving direction of the slider **40** and the second cutout face **42**CI extending in the direction substantially perpendicular to the moving direction of the slider **40**.

[0067] This can suitably avoid the unexpected release of the catching portion **41** from being caught on the caught portion **17** even if a distortion force occurs in the deflectable catching portion **41** before the temporary mating of the mating connector **200**.

[0068] In addition, if the feature of the present disclosure described above is provided, the feature is not limited to the configuration where, as shown in FIGS. **3**, **6**B, and the like, the long sides of the catching portion **41** of the slider **40** extend in the direction substantially perpendicular to the moving direction X of the slider **40**. For example, as shown in FIG. **14**, a configuration may be employed where the long sides of the catching portion **41**II of the slider **40**II extend in a direction substantially parallel to the moving direction X of the slider **40**II.

[0069] A mode of use of the connector **100** of the present disclosure will be described below.

[0070] First, the slider **40** is inserted through the second opening portion **15** of the outer housing **10** (see FIGS. **4**A, **4**B, and the like). Such insertion is performed until the deflective catching portion **41** provided at a predetermined spot in the slider **40** is enabled to be caught on the caught portion **17** of the outer housing **10**. The catching portion **41** of the slider **40**, with the operation lever **20** in the first position mentioned above, can be caught on the caught portion **17** of the outer housing **10**.

[0071] In addition, with the operation lever **20** pivotally supported on the cover **30** in the pre-turning first position, the cover **30** is attached to the outer housing **10**. In addition, the inner housing **50** is arranged in the outer housing **10** (see FIGS. **1**, **4**E, **9**, and the like).

[0072] As described in the above configuration section of the present disclosure, in the present disclosure, before the temporary mating of the mating connector **200**, the catching portion **41** of the slider **40** in the caught state on the caught portion **17** of the outer housing **10** is retained, so that the operation lever **20** can be retained in the pre-turning first position.

[0073] Also from the above, unless the temporary mating of the mating connector **200** is done, the abutment of the mating connector **200** against the catching portion **41** of the slider **40** and the subsequent deflection of the catching portion **41** to release the catching portion **41** of the slider **40** from the caught state do not occur.

[0074] Therefore, even if an external force to turn the operation lever **20** from the pre-turning first position toward the post-turning second position acts before the temporary mating of the mating connector **200**, the catching portion **41**, as shown in FIG. **10**, only shows a movement to abut against the body portion **42** of the slider **40** while the catching portion **41** of the slider **40** in the caught state on the caught portion **17** of the outer housing **10** is being retained. This enables the operation lever **20** to be retained in the pre-turning first position before the temporary mating of the mating connector **200**.

[0075] Next, with the inner housing **50** arranged, the enclosing portion **220** of the mating connector **200** is inserted into the continuous space **13** provided between the body portion **12** of the outer housing **10** and the inner housing **50**. By such insertion, the temporary mating of the mating connector **200** with the connector **100** of the present disclosure is performed.

[0076] In the present disclosure, the temporary mating of the mating connector **200** causes the mating connector **200** to abut against the catching portion **41** of the slider **40**, allowing the catching portion **41** to be deflected. This causes the catching portion **41** of the slider **40** to be released from the caught state on the caught portion **17** of the outer housing **10**.

[0077] Such release of the catching portion **41** of the slider **40** from the caught state during the temporary mating enables the operation lever **20** to be turned from the pre-turning first position toward the post-turning second position to slidingly move the slider **40** in the predetermined direction. Thereafter, when the sliding movement of the slider **40** is completed, the final mating of the mating connector **200** with the connector **100** of the present disclosure is performed.

[0078] Embodiments of the present disclosure have been described above, but the present disclosure is not limited to these, and various modifications based on knowledge of a person skilled in the art, such as combining the above configurations, may be made without departing from the spirit of the claims.

1. A connector adapted to mate with a mating connector, comprising:

an outer housing adapted to receive the mating connector, and including a catching portion;

a slider provided in the outer housing and slidingly movable in a predetermined direction the slider including a catching portion adapted to deflect at a predetermined position and be caught by the catching portion of the outer housing; and

an operation lever combined with the slider and adapted to be turned between a pre-turning first position and a post-turning second position, the slider is slidingly movable with turning of the operation lever, wherein the catching portion of the slider, with the operation lever in the first position, can be caught on the caught portion of the outer housing.

2. The connector according to claim **1**, wherein temporary mating of the mating connector causes the mating connector to abut against the catching portion of the slider to deflect the catching portion, thereby enabling the catching portion to be released from a caught state on the caught portion.

3. The connector according to claim **2**, wherein, before the temporary mating of the mating connector, the catching portion in the caught state on the caught portion is retained, and the operation lever is retained in the first position.

4. The connector according to claim **2**, wherein the release of the catching portion from the caught state during the temporary mating of the mating connector enables the operation lever to be turned from the first position toward the second position to slidingly move the slider in the predetermined direction.

5. The connector according to claim **2**, wherein the outer housing has a slider retaining space capable of retaining the slider therein.

6. The connector according to claim **5**, wherein the slider retaining space locally has, at an end side thereof, a catching portion accommodating space capable of accommodating the catching portion of the slider after the sliding movement is completed.

7. The connector according to claim **2**, wherein the deflectable catching portion has a protrusion at a distal-end side.

8. The connector according to claim **7**, wherein the outer housing has a slider retaining space adapted to retain the slider therein.

9. The connector according to claim **8**, wherein an inner wall portion of the outer housing forming part of the slider retaining space has an opening portion in which the protrusion is locatable.

10. The connector according to claim **2**, wherein the catching portion of the slider is provided inside a slit formed in a body portion of the slider.

11. A connector matable with a mating connector, comprising:

an outer housing capable of receiving therein the mating connector inserted thereinto; a slider provided in the outer housing and slidingly movable in a predetermined direction; and an operation lever combined with the slider and turnable between a pre-turning first position and a post-turning second position, wherein

the slider is slidingly movable with turning of the operation lever, and has a catching portion having deflectability at a predetermined position,

the outer housing has a caught portion on which the catching portion of the slider can be caught,

the catching portion of the slider, with the operation lever in the first position, can be caught on the caught portion of the outer housing, and

temporary mating of the mating connector causes the mating connector to abut against the catching portion of the slider to deflect the catching portion, thereby enabling the catching portion to be released from a caught state on the caught portion.

12. The connector according to claim **11**, wherein the catching portion in the caught state on the caught portion is retained and the operation lever is retained in the first position prior to the temporary mating of the mating connector.

13. The connector according to claim **11**, wherein the operation lever to be turned from the first position toward the second position to slidingly move the slider in the predetermined direction is enabled by the release of the catching portion from the caught state during the temporary mating of the mating connector.

14. The connector according to claim **11**, wherein the outer housing defines a slider retaining space adapted to retain the slider therein.

15. The connector according to claim **14**, wherein the slider retaining space includes a catching portion accommodating space adapted to accommodate the catching portion of the slider.

16. The connector according to claim **11**, wherein the deflectable catching portion has a protrusion at a distal-end side thereof, and the outer housing has a slider retaining space adapted to retain the slider therein.

17. The connector according to claim **16**, wherein an inner wall portion of the outer housing forming part of the slider retaining space defines an opening portion in which the protrusion is arranged.

18. The connector according to claim **11**, wherein the catching portion of the slider is arranged inside a slit formed in a body portion of the slider.

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